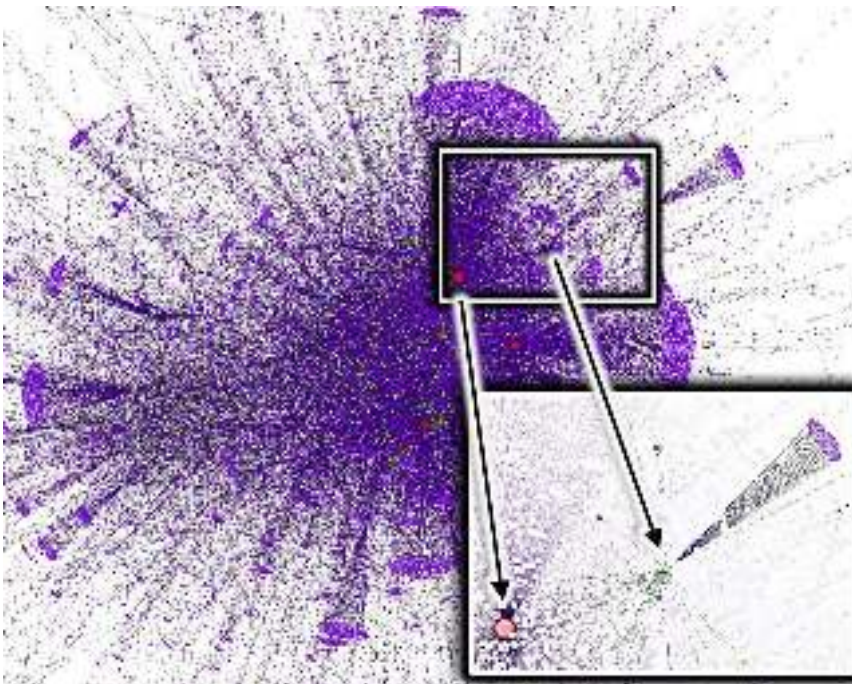
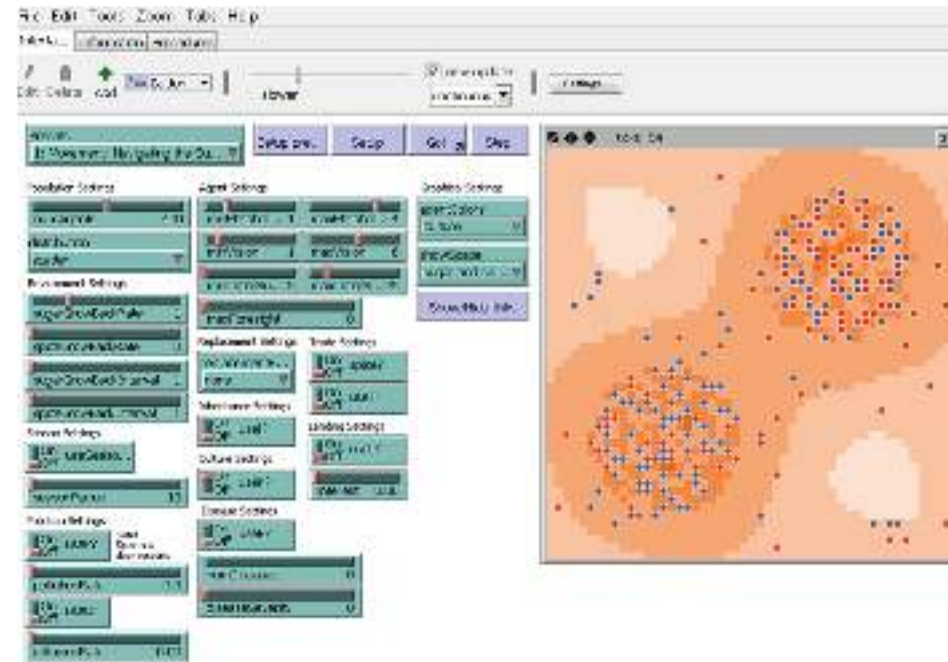




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Social Network Analysis (part 2): Network *Dynamics*



Martin Hilbert
Department of Communication
UC Davis

Today's questions

I. How do networks evolve?

II. How can we predict what kind of network will form?

- 1) Random networks
- 2) Scale-free networks
- 3) Small world networks
- 4) Hub & spoke / star networks...



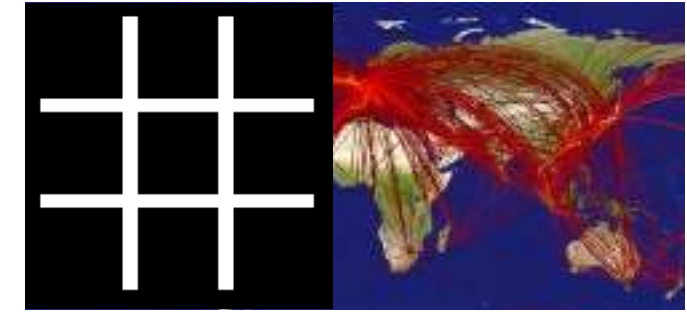
Growing *efficient* networks until they are *stable*

➤ Logic from (economic) cost & gains (leading to Nash equilibrium, Pareto optimality, etc.)

- Grow / form / rearrange network until it's "*in equilibrium*" = nothing changes
- ⇒ No node or link can be added or deleted (pairwise stable equilibrium)

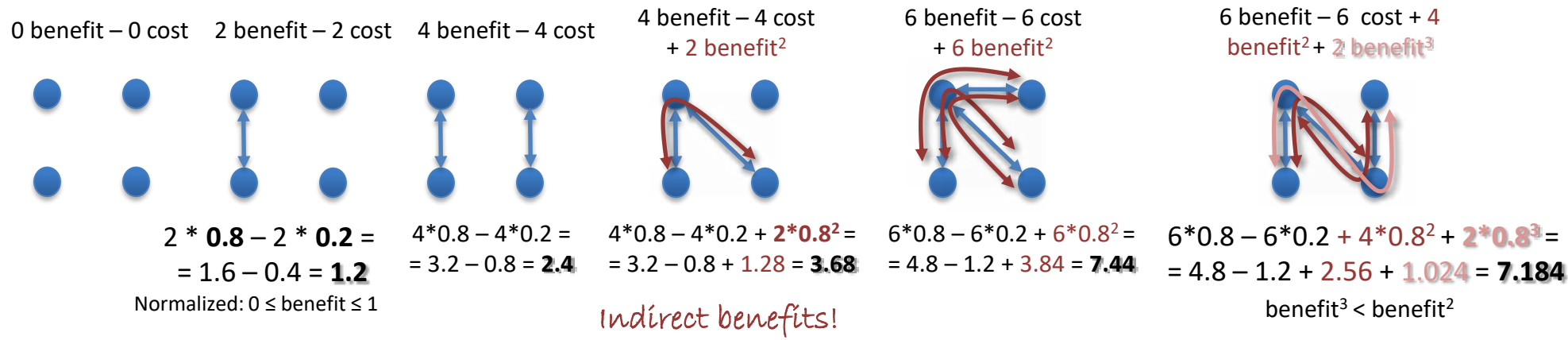
➤ Efficient Networks in the Symmetric Connections Model

- Given certain **cost** and **benefits of links**: what's the most efficient network structure?
 - **low cost** (benefit + benefit² > cost) => **complete** network is uniquely efficient
 - **medium cost** => **star** networks with all agents are uniquely efficient
 - **high cost** (benefit + (n-2)*benefit² < cost) => **empty** network is uniquely efficient



...because (accrd. to our assumptions)
shorter path length are more
valuable than longer ones

Why stars?



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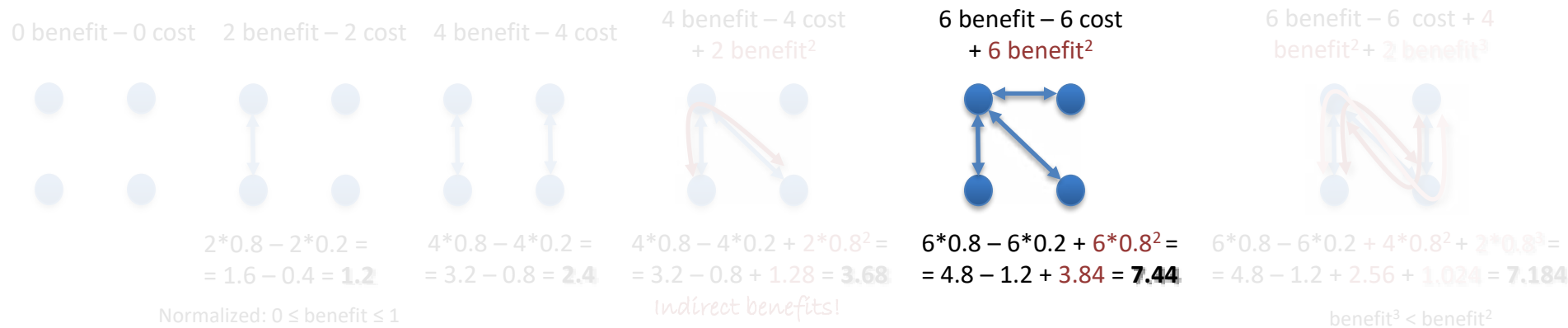


...because (accrd. to our assumptions) shorter path length are more valuable than longer ones



Why stars?

...BUT direct links are **ALSO more costly** than indirect links



Growing *efficient* networks until they are *stable*

$$1 \text{ benefit} - 1 \text{ cost} + 2 \text{ benefit}^2$$

$$=$$

$$1*0.8 - 1*0.2 + 2*0.8^2 = \mathbf{1.88}$$

$$6 \text{ benefit} - 6 \text{ cost}$$

$$+ 6 \text{ benefit}^2$$

$$3 \text{ benefit} - 3 \text{ cost} =$$

$$3*0.8 - 3*0.2 = \mathbf{1.8}$$

$$1 \text{ benefit} - 1 \text{ cost} + 2 \text{ benefit}^2$$

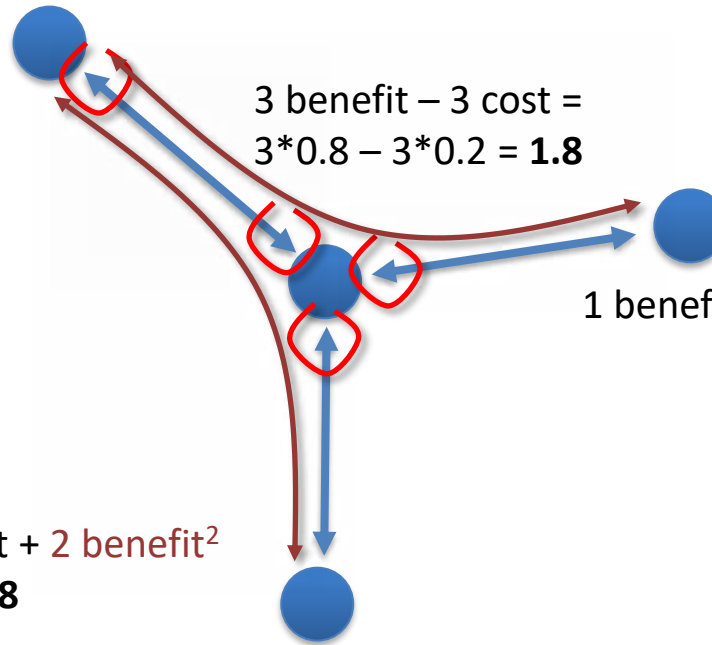
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$$6*0.8 - 6*0.2 + 6*0.8^2 =$$

$$= \mathbf{7.44}$$



...because (accord. to our assumptions)
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Strategic Network Intervention



$$1 \text{ benefit} - 1 \text{ cost} + 2 \text{ benefit}^2$$

$$=$$

$$1*0.8 - 1*0.2 + 2*0.8^2 = \mathbf{1.88}$$

$$- 0.02 = \mathbf{1.86}$$

$$6 \text{ benefit} - 6 \text{ cost} +$$

$$6 \text{ benefit}^2$$

$$3 \text{ benefit} - 3 \text{ cost} =$$

$$3*0.8 - 3*0.2 = \mathbf{1.8} + 0.02 + 0.02 + 0.02 = \mathbf{1.86}$$

$$1 \text{ benefit} - 1 \text{ cost} + 2 \text{ benefit}^2$$

$$= \mathbf{1.88}$$

$$- 0.02 = \mathbf{1.86}$$

$$1 \text{ benefit} - 1 \text{ cost} + 2 \text{ benefit}^2$$

$$= \mathbf{1.88}$$

$$- 0.02 = \mathbf{1.86}$$

$$6*0.8 - 6*0.2 + 6*0.8^2 =$$

$$= \mathbf{7.44} = \mathbf{4*1.86}$$

Strategic Network Intervention

Social efficiency:

sum of values of everybody is high

Social stability:

nobody can get better off by changing the network

But (!) doesn't always work:

- sometimes have to give to some without taking away from others
 - ⇒ requires outside resources (take away from unaffected...)
- sometimes have to take away from some without taking away from others
 - ⇒ accumulate extra resources (give to unaffected...)



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